



GeeksforGeeks

Data Science vs Software Engineering

Machine Learning Core Concepts



In Software Engineering, we know..



In Data Science, we know..



$$y = f(x)$$

$$\hat{y} \sim f(x)$$

DS vs AI vs ML vs DL vs Gen AI

Machine Learning Core Concepts



What is Data Science?

Data Science is the art and science of converting human intuition into algorithms and code.

Data Science and AI have become synonymous terms



What is Data Science?

Machine Learning uses traditional statistical models

Deep Learning uses neural networks

Gen AI uses Large Deep Learning Models

Agentic AI uses Large Language Models and Tools

Some Terms

Machine Learning Core Concepts



Dataset

The collected data

CSV, SQL, NoSQL, JSON, HTML, TXT PDF, Images, etc.

Each row/data point may be called an observation or instance or just row or data point.



Structured Dataset vs Unstructured Dataset vs Semi-Structured Dataset

Structured	Unstructured	Semi-Structured
Rigid Schema <i>Always fixed structure</i>	No Schema <i>No structure whatsoever</i>	Flexible Schema <i>Structure exists, but certain data points may break it</i>
Tabular	Multimodal	Text-based
CSV, SQL, Excel	Sentences, Image, Audio, Video	JSON, XML, Logs



Training vs Test Data

Train
80%

Used for training the model
The model learns patterns from the training data
Should be a representative sample

Test
20%

Used for evaluating the model AFTER training
Never shown to model during training



Features vs Target

Features = the input used to predict something

Also called independent variables, input, training features, etc.

Target = the something the model is trying to predict

Also called dependent variable(s), ground truth



Parameters vs Hyperparameters

Parameters are updated by models

Hyperparameters are provided by humans to the models



Categorical vs Numerical Data

Age is just a number. **Or, is it?**

Can it be negative?

Can it be a million?



Classification vs Regression

Regression = we are trying to predict a number

Ex: house prices, stock price, count of trees using satellite image, etc.

Classification = we are trying to predict a class (probability)

Ex: dog vs cat, lead propensity, next word, etc.

Standard ML Project Flow

Machine Learning Core Concepts



Business Problem Statement

Business Analyst

Gathers business requirements

Data Analyst

Understands existing data systems

Data Engineering

Applies transformations

Data Scientist

Builds models

Full Stack Engineer

Builds UI for models

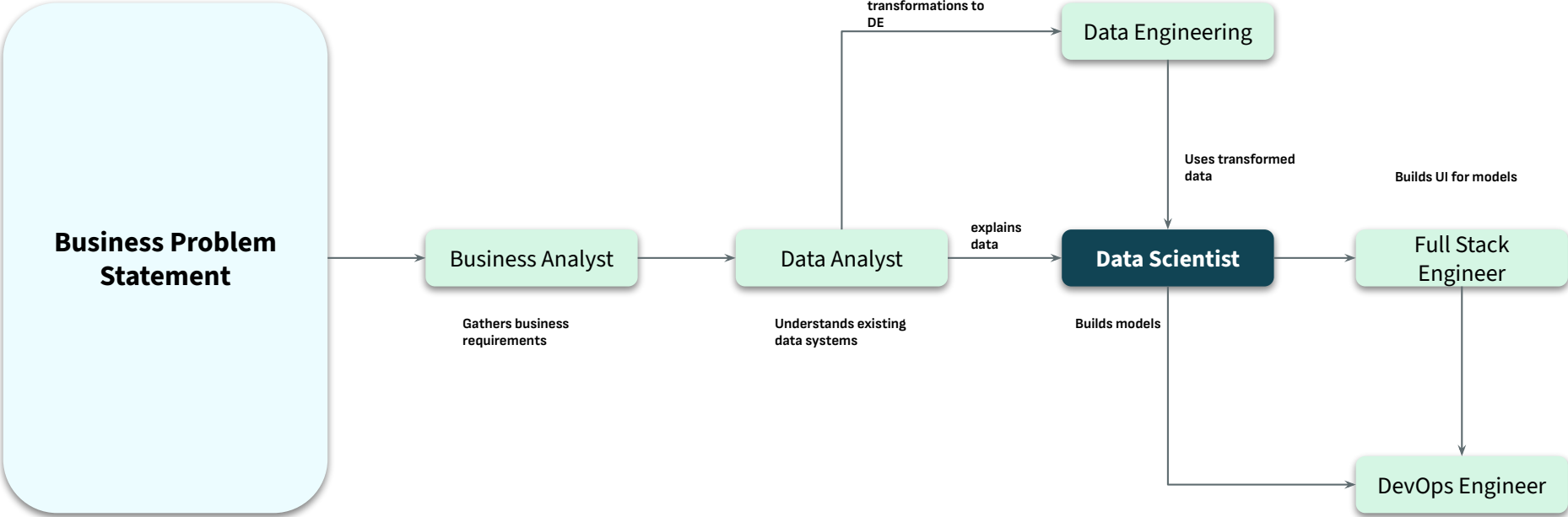
DevOps Engineer

Deploys UI and Models

explains transformations to DE

explains data

Uses transformed data



Different Types of Machine Learning

Machine Learning Core Concepts



Supervised Learning

We have both features and target.



Unsupervised Learning

We only have features, no target.



Reinforcement Learning

The model learns through trial and error using rewards and penalties.



Time Series Forecast

The previous inputs impact current output.



Anomaly Detection

Something seems out of the ordinary.



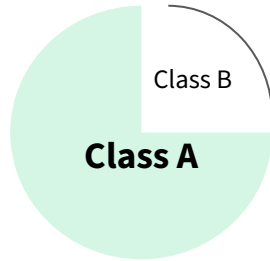
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Challenges of Machine Learning

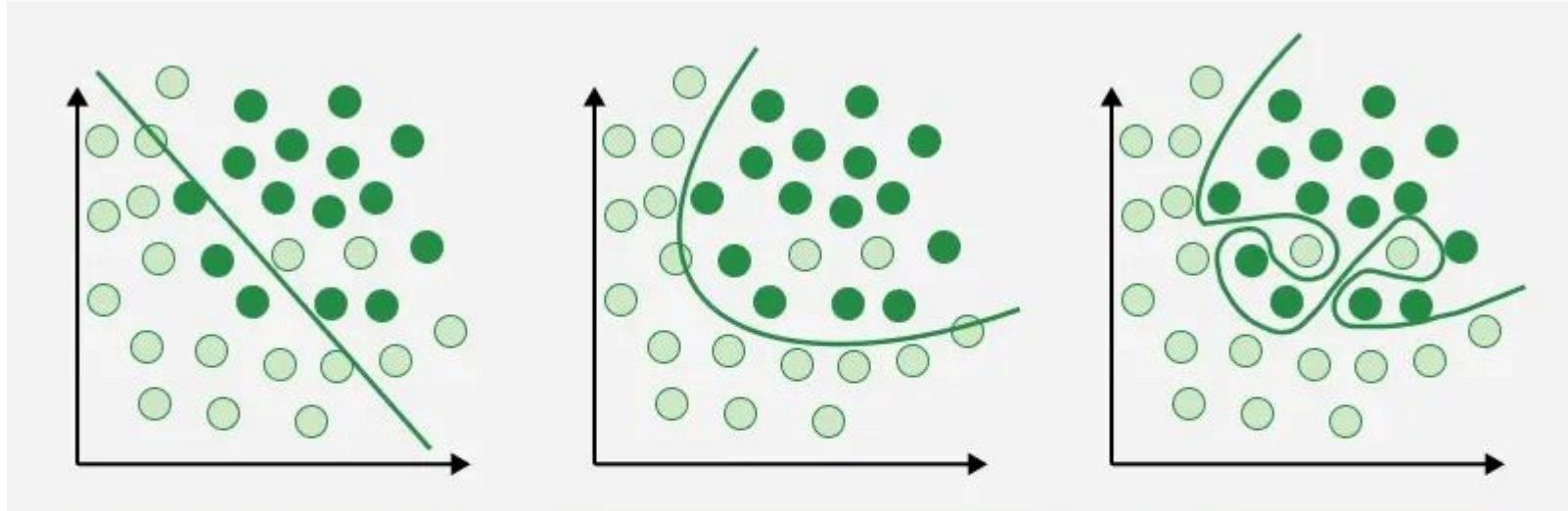
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Data Imbalance



Underfitting vs Overfitting

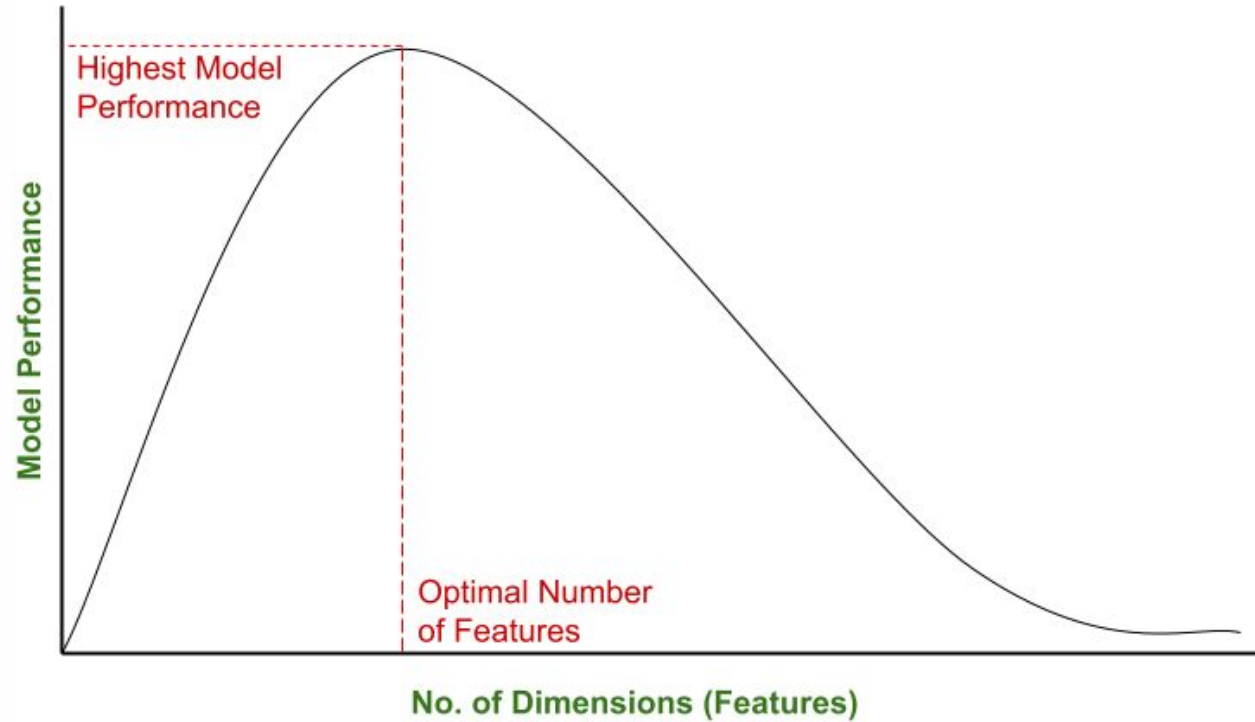


Underfitting

Appropriate

Overfitting

Curse of Dimensionality





Data Quality